



ITALIAN STATA CONFERENCE

Covariates in DiD

OLS specs, IPW, OR, DR

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Recall: the 2×2 — the two arithmetics

1. $\underbrace{[\mathbb{E}[Y|D=1, t] - \mathbb{E}[Y|D=1, t-1]]}_{\text{first difference}} - \underbrace{[\mathbb{E}[Y|D=0, t] - \mathbb{E}[Y|D=0, t-1]]}_{\text{first difference}}$
2. $\underbrace{[\mathbb{E}[Y|D=1, t] - \mathbb{E}[Y|D=0, t]]}_{\text{group difference}} - \underbrace{[\mathbb{E}[Y|D=1, t-1] - \mathbb{E}[Y|D=0, t-1]]}_{\text{group difference}}$

Difference over time, then groups. Or groups, then time. Same number.

Recall: the 2×2 — the four regressions

- 3. $Y = \alpha_1 + \alpha_2 \text{Post} + \alpha_3 D + \delta (\text{Post} \times D) + \varepsilon$ *saturated regression*
- 4. $Y = \alpha_1 + \tau_t + \sigma_i + \delta (\text{Post} \times D) + \varepsilon$ *two-way fixed effects*
- 5. $\Delta Y_i = \alpha_1 + \delta D_i + \varepsilon_i$ *first-difference / horizontal regression*
- 6. $\Delta \bar{Y}_G = \alpha_1 + \delta \text{Post}_G + \varepsilon_G$ *group / vertical regression*

All six are **numerically identical**.

The 2×2 DiD estimand

$$\underbrace{\hat{\delta}_{\text{DiD}}}_{\text{what we estimate}} = \underbrace{\text{ATT}}_{\text{what we want}} + \underbrace{\left[\mathbb{E}[\Delta Y(0) \mid D=1] - \mathbb{E}[\Delta Y(0) \mid D=0] \right]}_{\text{PT bias}}$$

Treatment = college education. Outcome = wages.
Parallel trends requires the PT bias term to equal zero.

A thought experiment

Group	Untreated wage trend $\Delta Y(0)$
Male workers	+10 per year
Female workers	+8 per year

Case 1: $D=1$ is 75% male. $D=0$ is 75% male.

Will parallel trends hold?

Case 1: show the work

	Calculation	Result
$\mathbb{E}[\Delta Y(0) \mid D=1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D=0]$	$0.75 \times 10 + 0.25 \times 8$	9.5

Parallel trends holds? **Yes.**

$$\hat{\delta}_{\text{DiD}} = \text{ATT} \quad \checkmark$$

Case 2: unequal composition

	% male
$D=1$ (college educated)	75%
$D=0$ (no college)	25%

Same wage trends as before (+10 male, +8 female).
Will parallel trends hold now?

Case 2: show the work

	Calculation	Result
$\mathbb{E}[\Delta Y(0) \mid D=1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D=0]$	$0.25 \times 10 + 0.75 \times 8$	8.5

Parallel trends holds? **No.**

$\hat{\delta}_{\text{DiD}} \neq \text{ATT}$ because PT bias = $9.5 - 8.5 = 1.0 \neq 0$

Why? Imbalance on a trend-causing covariate

DiD does **not** require the two groups to look the same on *levels*.

DiD **does** require the same $Y(0)$ *trends*.

Imbalance in a covariate that *causes* $Y(0)$ trends
will **mechanically** break parallel trends.

Fix: control for sex

If sex is the only covariate driving differential trends,
then *conditioning on sex* removes the PT bias.

Four ways to condition

1. TWFE saturated regressions
2. Inverse probability weighting (IPW)
3. Outcome regression (OR)
4. Doubly robust (DR)

We will discuss all four.

For conditioning to work: conditional PT

We need parallel trends *within* each group:

1. Male treated workers have the *same* $Y(0)$ trend as male control workers
2. Female treated workers have the *same* $Y(0)$ trend as female control workers

Conditional PTA: $\mathbb{E}[\Delta Y(0) \mid D=1, X] = \mathbb{E}[\Delta Y(0) \mid D=0, X]$ for all X

The standard TWFE with additive X

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \varepsilon_{it}$$

Just add in the covariates, then. Right?

Sure — if you're willing to make two assumptions you haven't stated.

Two hidden assumptions — and one fatal problem

1. Returns to X are **constant over time**
2. Treatment effect is **the same for all X**

Fatal problem: if X_i is time-invariant, it is collinear with α_i and drops out — whether you use unit fixed effects *or* a saturated dummy specification. An additive, separable X cannot deliver conditional parallel trends if the covariate is absorbed before it can do any work.

What if returns to X change over time?

Example: discrimination against women is not constant — $\beta_{\text{post}} \neq \beta_{\text{pre}}$.

Allow time-varying returns in the DGP:

$$Y_{it}(0) = \alpha_i + \gamma_t + \beta_t X_i$$

Write out the **post** and **pre** equations for unit i :

$$Y_{i,\text{post}}(0) = \alpha_i + \gamma_{\text{post}} + \beta_{\text{post}} X_i$$

$$Y_{i,\text{pre}}(0) = \alpha_i + \gamma_{\text{pre}} + \beta_{\text{pre}} X_i$$

Subtract, and a $\Delta\beta \cdot X$ term survives

Subtract (post – pre):

$$\Delta Y_i(0) = \underbrace{(\gamma_{\text{post}} - \gamma_{\text{pre}})}_{\Delta\gamma} + \underbrace{(\beta_{\text{post}} - \beta_{\text{pre}})}_{\Delta\beta} \cdot X_i$$

The trend in $Y(0)$ now *depends on* X . If the groups differ in X , their $Y(0)$ trends differ — and that is exactly a parallel-trends violation.

Now take expectations by group

Starting from $\Delta Y_i(0) = \Delta\gamma + \Delta\beta \cdot X_i$, condition on treatment status:

$$E[\Delta Y(0) \mid D=1] = \Delta\gamma + \Delta\beta \cdot E[X \mid D=1]$$

$$E[\Delta Y(0) \mid D=0] = \Delta\gamma + \Delta\beta \cdot E[X \mid D=0]$$

Parallel trends bias:

$$\underbrace{E[\Delta Y(0) \mid D=1] - E[\Delta Y(0) \mid D=0]}_{\text{PT bias}} = \Delta\beta \cdot (E[X \mid D=1] - E[X \mid D=0])$$

PT fails whenever **both**: returns shift ($\Delta\beta \neq 0$) **and** groups are imbalanced on X .

Fix for Bias #1: interact $\text{Post} \times X$

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \gamma (\text{Post}_t \times X_i) + \varepsilon_{it}$$

γ absorbs any shift in returns to X between pre and post.

(Spec B in the lab.)

Fix for Bias #1, in code

```
reg re i.post##i.ever_treated ///  
    i.post##c.age i.post##c.agesq i.post##c.agecube ///  
    i.post##c.educ i.post##c.educsq ///  
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///  
    i.post##c.re74 i.post##i.u74, robust
```

LESSON

Bias #2: Heterogeneous Treatment Effects

What if the effect of college differs by sex?

Previously we worried about **trends in** $Y(0)$ being unequal across groups.

Now the concern is different:

$$\mathbb{E}[Y(1) - Y(0) \mid \text{Male}] \neq \mathbb{E}[Y(1) - Y(0) \mid \text{Female}]$$

Do the rich and the poor get the same return to college?

Do workers in cities and towns get the same return?

If not, **and** the treatment group is imbalanced on these covariates,
the 2×2 DiD will not recover the ATT.

Proof, step 1 — the TWFE model and conditional PT

The TWFE model with one covariate X :

$$Y_{it} = \alpha_i + \gamma_t + \delta (T_t \times D_i) + \beta X_i + \varepsilon_{it}$$

Conditional parallel trends says, for each value of X :

$$\mathbb{E}[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid D=1, X] = \mathbb{E}[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid D=0, X]$$

Rearranged — the missing counterfactual for the treated is identified:

$$\mathbb{E}[Y(0)_{\text{post}} \mid D=1, X] = \mathbb{E}[Y_{\text{pre}} \mid D=1, X] + \mathbb{E}[Y_{\text{post}} \mid D=0, X] - \mathbb{E}[Y_{\text{pre}} \mid D=0, X]$$

Proof, step 2 — plug TWFE in: the slopes look the same

From the TWFE model, observed conditional means:

$$\mathbb{E}[Y_{\text{pre}} \mid D=1, X] = \alpha_1 + \alpha_3 + \beta X$$

$$\mathbb{E}[Y_{\text{post}} \mid D=0, X] = \alpha_1 + \alpha_2 + \beta X$$

$$\mathbb{E}[Y_{\text{pre}} \mid D=0, X] = \alpha_1 + \beta X$$

Counterfactual for treated (step 1 identity):

$$\mathbb{E}[Y(0)_{\text{post}} \mid D=1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \beta X$$

Treated outcome, post: $\mathbb{E}[Y(1)_{\text{post}} \mid D=1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta X$

Both potential outcomes carry the *same* β . Subtract: $\text{ATT} = \delta$. *Suspiciously clean.*

Proof, step 3 — what if $Y(1)$ and $Y(0)$ have different slopes in X ?

Let the slopes differ: write β_1 for the $Y(1)$ slope, β_2 for $Y(0)$:

$$\mathbb{E}[Y(1)_{\text{post}} \mid D=1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta_1 X$$

$$\mathbb{E}[Y(0)_{\text{post}} \mid D=1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \beta_2 X$$

True ATT given X :

$$\text{ATT}(X) = \delta + (\beta_1 - \beta_2) X$$

- If $\beta_1 \neq \beta_2$, the treatment effect *depends on X*
- TWFE fits one β , not two — so $\hat{\delta}$ recovers ATT only if $\beta_1 = \beta_2$

Proof, step 4 — TWFE imposes homogeneous δ in X

Assumption: TWFE-with- X requires the treatment effect to be the same at every value of X .

- If X is **sex**: effect must be equal for men and women
- If X is **income**: effect must be equal at \$1 and at \$1,000,000
- If X is **age**: effect must be equal at 25 and at 65

OLS doesn't warn you that you're making this assumption. The math does.
If effects actually vary with X , TWFE averages across the variation
and returns a weighted parameter you didn't ask for.

Fix for Bias #2: saturate treatment with X

1. **Run a saturated regression on first differences** — interact treatment with every X
2. **Predict $\Delta Y(1)$ and $\Delta Y(0)$ for each unit** — average the gap in the treated group

Interacting D with every X lets the treatment effect vary with X — which is exactly what additive X forbade.

Fix for Bias #2, in code

```
* 1. Saturated regression on first differences
reg dy i.ever_treated ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.educsq ///
    i.ever_treated#i.marr i.ever_treated#i.nodegree ///
    i.ever_treated#i.black i.ever_treated#i.hisp ///
    i.ever_treated#c.re74 i.ever_treated#i.u74 ///
    $X, robust

* 2. Predict potential outcomes; average the gap in the treated group
replace ever_treated = 1
quietly predict dy_hat_1, xb
replace ever_treated = 0
quietly predict dy_hat_0, xb
replace ever_treated = evt_orig // restore
gen tau_sat = dy_hat_1 - dy_hat_0
quietly sum tau_sat if ever_treated == 1 // ATT = r(mean)
```

Bias #3: time-invariant covariates get absorbed

If you need X_i for conditional PT, but X_i doesn't change over time,
TWFE will **delete it**.

TWFE “demeans” the data. The mean of a constant is itself.
So $X_i - \bar{X}_i \equiv 0$ — it drops out of the regression entirely.

Fix: include $X_i \times \text{Post}_t$ instead. The Post interaction is time-varying,
so it survives demeaning and can do the work of conditioning on X .

Three biases — one family of fixes

Bias	What TWFE imposes	Fix
#1 Trend imbalance	β constant pre and post	$\text{Post} \times X$
#2 Heterogeneous TE	δ same at every X	$D \times X$ (saturate treatment)
#3 Time-invariant X	X_i survives demeaning	$\text{Post} \times X$ (same fix as #1)

But if you remember only two

1. Imbalance bias

The groups differ in an X whose returns move over time, so their $Y(0)$ trends differ.

2. Heterogeneous treatment effect bias

The effect varies with X , and additive TWFE forbids that.

Bias #3 is real but mechanical — and it takes the same fix as #1.

One thing we are not covering: covariates measured *after* treatment

Everything here assumes X is measured at **baseline**.

Post-treatment covariates are their own problem,
and they have their own paper.

Caetano and Callaway, forthcoming, Review of Economics and Statistics.

What that means in practice

Most researchers include additive X and call it a day. That is the wrong spec.

To fix all three you need *both* $\text{Post} \times X$ and $D \times X$.

It is rare to see this in published work.

LESSON

Alternative Estimators

Three more estimators

Now we will look at three more estimators.

These are alternatives to the regression-based specs we just covered.

1. **IPW** — reweight controls to look like treated
2. **Outcome regression (OR)** — model the control trend, impute the counterfactual
3. **Doubly robust (DR)** — combine both; consistent if *either* is right

IPW: reweight the controls to look like the treated

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)} \Delta Y_i$$

$$\hat{p}(X_i) = \widehat{\text{Pr}}(D_i = 1 \mid X_i) \quad (\text{propensity score}) \quad \Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$$

IPW: five steps

1. **Estimate $\hat{p}(X)$.** Logit of D_i on X_i .
2. **Check overlap.** Plot $\hat{p}$ by treatment status. Trim if needed.
3. **Construct weights.** $w_i = (D_i - \hat{p}(X_i)) / (1 - \hat{p}(X_i))$.
4. **Weighted average.** $\widehat{ATT} = \sum w_i \Delta Y_i / n_T$.
5. **Standard error.** Bootstrap or influence function.

Overlap is non-negotiable

If $\hat{p}(X_i) \rightarrow 1$ for a control unit, its weight $\rightarrow \infty$.

No overlap = no valid counterfactual.

This is a data problem. No estimator fixes it.

OR — or “regression adjustment”

$$\widehat{\text{ATT}}_{\text{OR}} = \frac{1}{n_T} \sum_{D_i=1} \left[\Delta Y_i - \hat{\mu}_0(X_i) \right]$$

$$\hat{\mu}_0(X_i) = \hat{E}[\Delta Y \mid D=0, X] \quad \text{estimated on controls only}$$

Same estimator, newer name

You will increasingly see this called
regression adjustment (RA).

Same idea. Newer name. Heckman, Ichimura and Todd got there in 1997.

OR: five steps

1. **First-difference.** $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$.
2. **Fit outcome model.** Regress ΔY on X *using controls only*.
3. **Predict counterfactual.** Evaluate $\hat{\mu}_0(X_i)$ for each treated unit.
4. **Mean gap.** $\text{ATT} = \overline{\Delta Y}^T - \overline{\hat{\mu}_0(X)}^T$.
5. **Standard error.** Bootstrap.

Surprise — you already know OR

OR step 2: *“Regress ΔY on X using controls only.”*

That is exactly what the saturated TWFE does.

Let's prove it slowly.

Proof, step 1 — the saturated model

Spec C runs this regression on first differences:

$$\Delta Y_i = \underbrace{\alpha}_{\text{intercept}} + \underbrace{\delta D_i}_{\text{treatment}} + \underbrace{\beta' X_i}_{\text{covariates}} + \underbrace{\tau' (D_i \times X_i)}_{\text{treatment interactions}} + \varepsilon_i$$

$D_i \in \{0, 1\}$ and **every** element of X_i is interacted with D_i .

Proof, step 2 — separate by group

Plug $D_i = 0$ and $D_i = 1$ into the model:

$$D_i = 0 : \quad \Delta Y_i = \alpha + \beta' X_i + \varepsilon_i$$

$$D_i = 1 : \quad \Delta Y_i = (\alpha + \delta) + (\beta + \tau)' X_i + \varepsilon_i$$

The two groups have **entirely different** parameters.

Controls contribute to (α, β) . Treated contribute to $(\alpha + \delta, \beta + \tau)$.

Proof, step 3 — OLS separates completely

Re-label: $a_0 = \alpha$, $b_0 = \beta$, $a_1 = \alpha + \delta$, $b_1 = \beta + \tau$.

The OLS objective is:

$$\min_{a_0, b_0, a_1, b_1} \underbrace{\sum_{D_i=0} (\Delta Y_i - a_0 - b'_0 X_i)^2}_{\text{only controls}} + \underbrace{\sum_{D_i=1} (\Delta Y_i - a_1 - b'_1 X_i)^2}_{\text{only treated}}$$

The two sums share **no parameters**.

$(\hat{a}_0, \hat{b}_0) = \text{OLS of } \Delta Y \text{ on } X \text{ **using controls only** .}$

Proof, step 4 — Spec C and OR make the same imputation

For a treated unit i , what does each estimator predict as $\Delta Y_i(0)$?

Spec C (saturated TWFE)	set $D_i=0$:	$\hat{a}_0 + \hat{b}'_0 X_i$
OR / regression adjustment	control OLS:	$\hat{a}_0 + \hat{b}'_0 X_i$

Identical prediction $\Rightarrow$ identical ATT. **Spec C is OR.**

One estimator, two names — just like the 2×2

Recall: six formulas for the 2×2 , all giving the same number.

Same principle here.

Saturated TWFE (Spec C) and outcome regression (OR / RA)
are **the same estimator** expressed two different ways.

We will confirm this in the do file: both print \$1,770.

DR: two chances to be right

$$\widehat{ATT}_{DR} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{\left[\Delta Y_i - \hat{\mu}_0(X_i) \right]}_{\text{OR residual}}$$

DR is consistent if *either* model is right — not both

$\hat{\mu}_0$ correct	$\hat{p}$ correct	Consistent?
✓	✓	✓
✓	×	✓
×	✓	✓
×	×	No

Read that last row

“Doubly robust” $\neq$ doubly unbeatable.

Two chances to be right is not the same as a guarantee.

Get both models wrong and DR is wrong too.

Lab: ten estimators, one benchmark

Labs/Lalonde-Covariates/lalonde_all_specs.do

NSW treated + CPS controls · Pre = 1975, Post = 1978

RCT benchmark: **\$1,794**

Spec 0 (Naive) · Spec A (Additive X) · Spec B (Post $\times X$) · Spec C (Saturated)

OR by hand · OR drdid · IPW by hand · IPW drdid · DR by hand · DR drdid

A Monte Carlo to rank the estimators

Estimators: TWFE, OR, IPW, DR

DGPs: 4 (vary whether OR model and pscore are correct)

n : 1,000 Replications: 10,000

Report: Bias, RMSE, SE, coverage, CI length

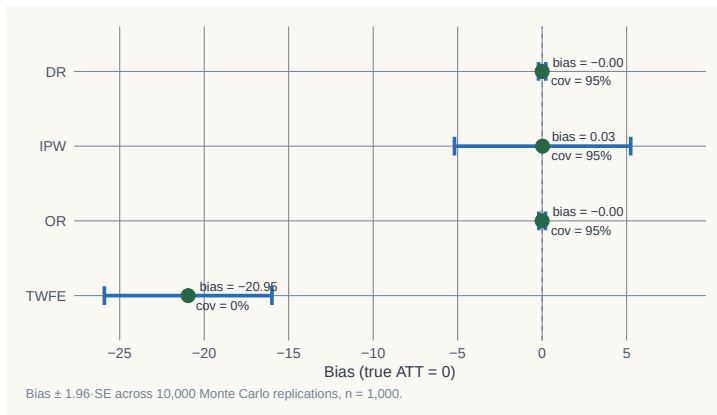
When does DR pay off? When does it break?

DGP 1: both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	−20.95	21.12	2.53	0.000	9.91
OR	−0.001	0.10	0.10	0.950	0.40
IPW	+0.026	2.77	2.66	0.952	10.44
DR	−0.001	0.11	0.11	0.947	0.41

TWFE: coverage zero. OR and DR: tight on truth. IPW: unbiased but wide.

DGP 1: the sampling distribution

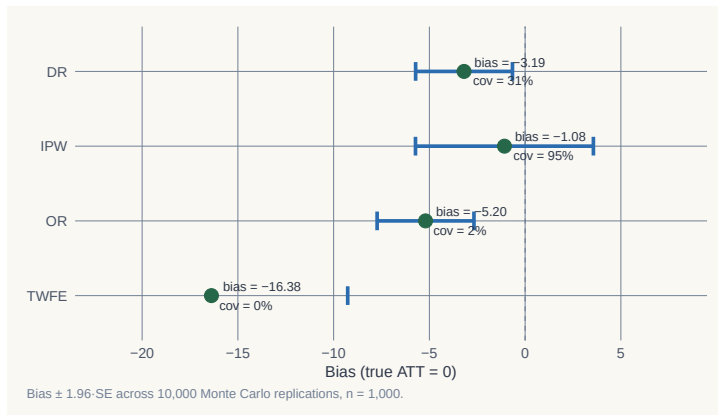


DGP 4: both models wrong

	Bias	RMSE	SE	Coverage	CI length
TWFE	−16.38	16.54	3.63	0.000	14.22
OR	−5.20	5.36	1.29	0.015	5.05
IPW	−1.08	2.66	2.37	0.949	9.31
DR	−3.19	3.45	1.29	0.308	5.07

DR undercovers at 0.31. IPW wins. Doubly robust $\neq$ doubly unbeatable.

DGP 4: the sampling distribution



What the Monte Carlo teaches

- **TWFE + additive X** : coverage zero in every DGP. **Stop using it as a default.**
- **OR**: efficient when right. Coverage 1.5% when wrong.
- **IPW**: less efficient, but robust to outcome misspecification.
- **DR**: best when one model is right. No better than IPW when both are wrong.

Adding X doesn't make your DiD credible

Adding X shifts the credibility question—
from **“does PT hold?”**
to **“is my model of $Y(0)$ or $p(X)$ correct?”**

You still have to answer it.

Now let's see it with real data

LaLonde (1986) · Dehejia & Wahba (2002)

A famous dataset where we **know the true effect**.

The benchmark: **\$1,794**.

LaLonde (1986): the RCT found $\approx \$800$

- Job training program (National Supported Work Demonstration)
- Randomized — treated and control groups assigned by lottery
- RCT estimate: earnings effect $\approx \$800$

A clean causal estimate. The benchmark exists.

LaLonde: dropped the RCT controls, used CPS instead

- Replaced the randomized controls with observational data (CPS, PSID)
- Re-ran every evaluator method from the literature
- Results ranged from **−\$16,000 to +\$3,000**

The true effect was \$800. Not one method recovered it.

“The available evidence calls into serious question the ability of nonexperimental evaluators. . . ”

Empirical labor economics faces an evaluation problem.

Dehejia & Wahba (2002): propensity scores get close

- Need **two pre-treatment years** for all datasets — uses a shorter sample
- RCT benchmark on this shorter sample: **\$1,794**
- Applied propensity score matching on the non-experimental sample
- Result: estimates close to \$1,794

Selective use of the data + right method = nearly recovered the truth.

Our exercise: DiD with and without covariates

NSW treated + CPS controls · Pre = 1975 · Post = 1978

We know the effect: **\$1,794**

Let's see which specs recover it — and which don't.

Naive TWFE: \$3,621 — almost double the truth

```
reg re i.post##i.ever_treated, robust
```

Estimator	ATT	vs benchmark
Naive TWFE	\$3,621	+\$1,827

No covariates — the CPS controls trend very differently from NSW treated.

Additive X : still \$3,621

```
reg re i.post##i.ever_treated $X, robust
```

Estimator	ATT	vs benchmark
Additive X	\$3,621	+\$1,827

Adding X as time-invariant regressors does nothing here. The unit FE absorbs the level differences. The trend bias remains.

Post $\times$ X: \$1,711 — now in the right range

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.agecube ///
    i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree ///
    i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##i.u74, robust
```

Estimator	ATT	vs benchmark
Post $\times$ X	\$1,711	-\$83

Saturated TWFE: \$1,770

```
* First-difference, then regress dy on X + D x X interactions
reg dy $X ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.re74 ///
    i.ever_treated, robust
* Predict counterfactuals; ATT = mean(tau_sat) for treated
```

Estimator	ATT	vs benchmark
Saturated TWFE	\$1,770	−\$24

OR (HIT 1997): same answer as saturated TWFE

```
* Fit trend model on controls only
reg dy $X if ever_treated == 0, robust
predict dy_hat
* ATT = mean gap for treated
gen delta_hat = dy - dy_hat if ever_treated == 1
```

Estimator	ATT	vs benchmark
OR by hand / drdid reg	\$1,770	−\$24

Saturated TWFE and OR are algebraically equivalent in the 2-period case.

IPW (Abadie 2005): \$1,861

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

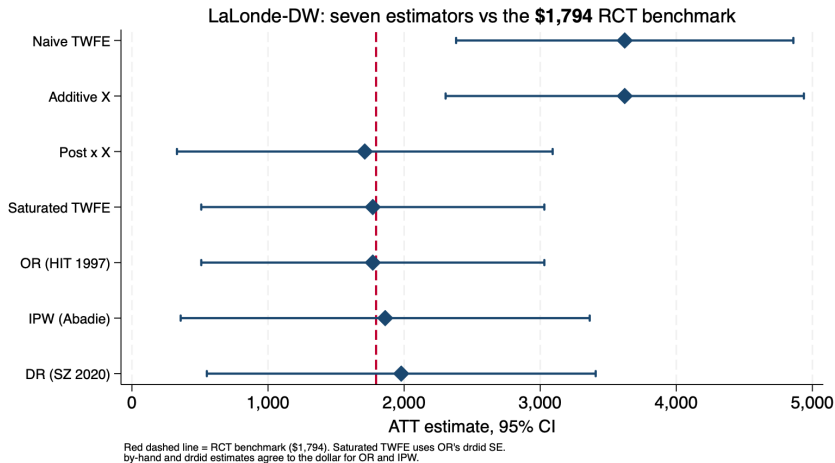
Estimator	ATT	vs benchmark
IPW	\$1,861	+\$67

DR (Sant'Anna–Zhao 2020): \$1,979

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

Estimator	ATT	vs benchmark
DR	\$1,979	+\$185

All seven estimators vs the \$1,794 benchmark



Covariates are not robustness checks — they are design

- People say: *“if adding covariates changes results, be suspicious.”*
- That gets it backwards.
- We **know** the effect is \$1,794.
- Without covariates, or with additive covariates, we get \$3,621 — *wrong*.
- The correct specs recover \$1,770–\$1,979 — *right*.

Covariates fix broken parallel trends. Changing your result is not a bug. It is the design working.

LESSON

In Conclusion

How to choose covariates, and how to check them

This is not omitted variable bias

The instinct is: *“I forgot a variable, so I’ll put it back in.”*

That instinct comes from cross-sectional regression.
It is the wrong instinct here.

A covariate here describes a *population*

X is not a missing regressor.

X marks **groups of people whose untreated outcomes were going to move differently anyway.**

And it is $Y(0)$, not Y

The object is $\Delta \mathbb{E}[Y(0) \mid X]$

Not the level. Not the observed outcome.

The *trend* in the *untreated potential outcome*, within a group defined by X .

Our example, in those terms

Men and women had different earnings trajectories
whether or not anyone was treated.

Sex did not “cause” the outcome and get left out of a regression.

Sex labelled two populations with different $\Delta Y(0)$.

The groups had different mixes of them. That is the whole problem.

So the common-sense method for choosing X

Ask one question about each candidate covariate:

“Would this group’s $Y(0)$ have trended differently?”

If yes, it belongs. If it only predicts the *level*, it does not.

Then two things to check

1. **Balance** — are the groups actually different in X ?
2. **Overlap** — can you find comparable controls at all?

1. Balance: the standardized difference in means

$$\frac{\bar{X}_1 - \bar{X}_0}{\sqrt{\frac{1}{2}(V_1 + V_0)}}$$

The gap between group means, scaled by the pooled standard deviation.
Unit-free, so it is comparable across variables.

The threshold

$$\left| \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{\frac{1}{2}(V_1 + V_0)}} \right| > \mathbf{0.25}$$

If you have that, it is **imbalanced**.

Imbens and Rubin (2015).

And then including it is not optional

Imbalanced + predicts $\Delta Y(0)$ = **you must condition on it**

Our sex example cleared both bars. That is exactly why the naive 2×2 was wrong.

2. Estimate the propensity score

$$p(X) = \Pr(D = 1 \mid X)$$

One number per unit: how treated-looking is this observation?

Then plot it — two histograms, one axis

Plot the distribution of $p(X)$ **separately for treatment and control.**

This is the single most informative picture in the whole covariate discussion,
and almost nobody shows it.

Look for overlap

Do the two distributions **sit on top of each other?**

Where they overlap, you have comparable controls.

Where they don't, you are extrapolating.

If the tails misbehave: **consider trimming the control group.**

The IPW weight

$$\frac{p(X)}{1 - p(X)}$$

Applied *only* to the comparison group.

For the ATT, the treated units' propensity score is never used

DiD estimates the **ATT**.

The treated group is already the population we want.

We do not reweight it. We reweight the controls *to look like it*.

Where that comes from

$$\mathbb{E} \left[\left(\frac{D}{\mathbb{E}[D]} - \frac{(1-D) \frac{p(X)}{1-p(X)}}{\mathbb{E} \left[(1-D) \frac{p(X)}{1-p(X)} \right]} \right) \Delta Y \right]$$

Two terms. Look at what each one does.

When $D = 1$

$$\mathbb{E} \left[\left(\frac{D}{\mathbb{E}[D]} - \frac{\overbrace{(1-D)}^{=0} \frac{p(X)}{1-p(X)}}{\mathbb{E} \left[(1-D) \frac{p(X)}{1-p(X)} \right]} \right) \Delta Y \right]$$

The second term carries $(1 - D)$. **It vanishes.**

Treated units enter with weight $1/\mathbb{E}[D]$.

Their propensity score is nowhere in the expression.

When $D = 0$

$$\mathbb{E} \left[\left(\frac{\overbrace{D}^{=0}}{\mathbb{E}[D]} - \frac{(1-D) \frac{p(X)}{1-p(X)}}{\mathbb{E} \left[(1-D) \frac{p(X)}{1-p(X)} \right]} \right) \Delta Y \right]$$

The first term carries D . **It vanishes.**

Control units enter weighted by $p(X)/(1-p(X))$.

That is the IPW, and it applies to them alone.

Now watch what happens as $p(X) \rightarrow 1$

$$\frac{p(X)}{1 - p(X)} \longrightarrow \infty$$

A control unit that *looks* treated gets **enormous** weight.

One observation can swamp the estimate. **This is why we trim.**

3. A different problem: perfect separation

Trimming fixes a *tail*.

What if there is **no overlap at all?**

Where separation comes from

- **Overfitting.** Enough covariates and a flexible enough model will separate anything.
- **Genuinely selective X .** Selection on observables is so sharp that little randomness is left.

The first is a modelling mistake. The second is a fact about your design.

If it's real, you are near an RDD

Regression discontinuity has **no overlap either** — by construction.

Nobody just above the cutoff is comparable to anybody just below,
except in the limit. That is the whole design.

So IPW cannot help you

Abadie (2005) needs **common support**.

Under separation you do not have it —
and no amount of trimming creates it.

Reach for outcome regression instead

Heckman, Ichimura and Todd (1997).

Model the control trend, then *impute* the missing counterfactual for the treated units.

But understand what you just took on

You are **imputing**.

The answer is now **highly dependent on functional form**.

IPW let the data pick comparable controls. OR makes you assert a shape.

Which means: be flexible, then be careful

- Use a **highly flexible specification** — machine learning is reasonable here.
- Flexible models overfit. If yours does, consider **penalized regression**.

You are trading a support problem for a specification problem.
Do it with your eyes open.

The checklist, in one place

1. Choose X that plausibly drives $\Delta Y(0)$ — *populations*, not omitted variables
2. Check balance: standardized difference $> |0.25|$ means imbalanced
3. Estimate $p(X)$; plot it by group; look for overlap
4. Trim the control tail where $p(X) \rightarrow 1$
5. If there is no overlap at all, stop — IPW will not save you
6. Use OR with a flexible spec, and penalize if it overfits

More on this in Causal Inference: The Remix, the unconfoundedness chapter — where I work through the Imbens-inspired checklists.

The question is never

“did I control for X ?”

The question is always

“did I get the model right?”
